Design Documentation for Website Advertising Integration Tool

1. Introduction

This document describes the design of a tool developed to integrate Google Ads API functionalities for the websites DiscussionSpot.com and GSMSharing.com. The purpose of the tool is to automate ad performance monitoring, keyword insights, and campaign optimization.

2. System Overview

The integration tool connects the websites with Google Ads API using secured OAuth2 authentication. The tool retrieves campaign data, keyword insights, and user engagement metrics and displays aggregated analytics.

3. Architecture

- Frontend: Dashboard built in ASP.NET Core MVC.

- Backend: C# service using Google Ads API SDK.

- Data Storage: MSSQL for logging and historical insights.

- Authentication: OAuth2 Client ID, Client Secret, Developer Token, and Refresh Token.

- API Calls: CampaignService, KeywordPlanService, GoogleAdsService.

4. Core Features

- Automatic retrieval of campaign performance metrics.

- Keyword suggestions and traffic forecast.

- Custom alerts for CPC changes or low performance.

- Exportable reports in CSV and PDF formats.

- Secure credential management via google-ads.yaml.

5. Workflow

1. User logs into admin dashboard.

2. Backend loads Google Ads API credentials.

3. App requests updated metrics.

4. Results are displayed and optionally saved in database.

5. User may generate and download reports.

6. Security Considerations

- OAuth tokens stored encrypted.
- No direct exposure of developer token.
- Restricted API scopes.
- HTTPS enforcement.

7. Future Enhancements

- AI-assisted keyword recommendations.
- Automated ad budget adjustments.
- Multi-account integration.

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